CLAUSESHIFT

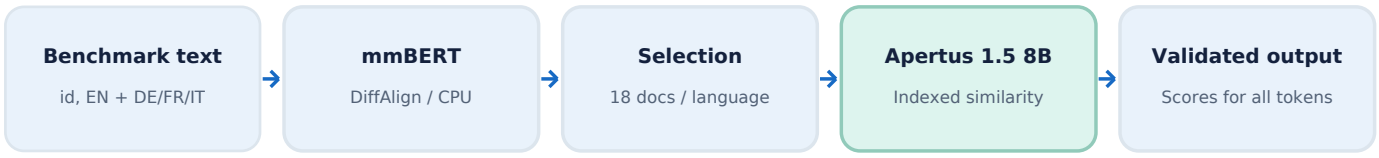
Evidence-grounded, word-level cross-lingual semantic difference recognition

Short method description | Guelmbaye Firmin DJIKOLOUM | 16 October 2026

Task and objective

For each English administrative document paired with its German, French or Italian counterpart, CLAUSESHIFT assigns a semantic difference score to every whitespace-delimited token on both sides. The system preserves original document identifiers, token order and punctuation-masking conventions. The competition metric is the macro average of token-level Spearman correlations over EN–DE, EN–FR and EN–IT.

Method at a glance



1. Public multilingual baseline. The frozen pipeline uses SwissGov-RSD DiffAlign with public jhu-clsp/mmBERT-base, last transformer layer (layer = −1), to score the complete bilingual dataset. No annotated final labels are used for training or inference.
2. Deterministic, bounded Apertus contribution. Without reference labels, length-stratified selection picks 18 documents per language (54 total). Baseline differences identify paired 32-word windows; up to 12 token indices per side are sent to swiss-ai/apertus-v1.5-8b through an OpenAI-compatible endpoint. The model returns explicitly indexed similarity grades in [0, 5].
3. Conservative score fusion. Only valid Apertus responses update selected tokens:
   difference = 0.65 × mmBERT_difference + 0.35 × (1 − Apertus_similarity / 5).
   All other tokens retain baseline scores. Invalid API responses fail closed; at most one remote attempt is made per selected document, with no automatic retry.
4. Long-document coverage. V1.15.2 applies overlapping bilingual word windows when a document exceeds 3,072 subtokens. Each inference remains within the cap, with 24 contextual words around each core passage. Core scores are stitched back to exact original token positions; missing or misaligned output triggers a hard failure rather than truncation.

Reproducibility safeguards

SwissGov-RSD source revision 1807a42100e7..., mmBERT revision c5955035435e..., Docker/Python 3.12, deterministic plans and signed checkpoints are pinned or recorded. Final inputs are reduced to id, text_a, text_b before inference. Structural validation checks ID/order, token coverage, numeric ranges and finiteness. The freeze manifest identifies the submission method as H1-HYBRID (digest prefix 2d5d89cc996e).

Evaluation & delivery

Measured evidence, final artifacts and reproducibility constraints

Development evaluation (not hidden-test evaluation)

Method / sample	EN–DE	EN–FR	EN–IT	Macro
Public mmBERT, full DEV (168/lang)	0.146786	0.087821	0.152947	**0.129185**
mmBERT, 18 selected DEV/lang	0.126881	0.065680	0.127844	**0.106802**
Hybrid, same selected sample	0.128646	0.063163	0.131595	**0.107801**
Hybrid minus baseline, selected	+0.001766	−0.002517	+0.003751	**+0.001000**

The baseline was scored on all 504 DEV documents using the official evaluator. The H1 ablation covers 54 selected DEV documents, not full DEV: 44/54 accepted Apertus responses, 10 fallbacks and 1,056 changed token scores. Its macro gain is only +0.001000 on that selected sample; French performance decreased. No statistically robust improvement, full-DEV hybrid score or hidden-test Spearman is claimed.

Final deliverables and integrity

The frozen Docker runner completed 224 documents each for EN–DE, EN–FR and EN–IT (672 documents total). An independent offline validation confirmed original document order, IDs, token counts and in-range scores for all three outputs. The resulting files and SHA-256 hashes are:

EN–DE: CLAUSESHIFT_admin_de.jsonl.jsonl — 97b81c8b33a0be09…

EN–FR: CLAUSESHIFT_admin_fr.jsonl.jsonl — a97df1757ceb1513…

EN–IT: CLAUSESHIFT_admin_it.jsonl.jsonl — 320692e348029c23…

These are structural integrity results, not a hidden-set quality score. The final gold labels were not used to optimize or evaluate the submitted predictions.

Execution and limitations

Executed CPU-only on Ubuntu 24.04 with 8 Intel vCPUs and 16 GiB RAM using Docker. A fresh checkout requires internet access to pinned source/model artifacts and an API key for https://api.publicai.co/v1 (LLM_NAME, LLM_BASE_URL, LLM_API_KEY). From the repository root, make run performs model warmup and execution. External API serving and responses are not fully controlled and may vary across fresh runs. Long-input proportional segmentation approximates cross-language boundaries and can affect scores.

Disclosure of freeze change. V1.15.1 correctly halted on an over-cap final input. V1.15.2 added the general segmentation method above and was re-frozen. This change was motivated by completeness, not by consulting final gold labels or tuning final-set performance. Admissibility of a post-first-access completeness fix remains subject to organizer rules.

Repository and provenance

github.com/guelmbaye/clauseshift · track_2a/technical_report.md · Apache-2.0 code license. The source identifies upstream third-party artifacts and their respective licenses.